

HIGHER NATIONAL DIPLOMA IN INFORMATION TECHNOLOGY

ADVANCED TECHNOLOGICAL INSTITUTE GALLE



2nd year 2nd Semester Individual Project Proposal

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INTRODUCTION

The proposed system is a QR code–based Student Management and Attendance System developed for an educational institute called **Sigma Institute**. The main goal of this system is to improve the way student information, attendance, and fee payments are managed by making the whole process more digital, accurate, and efficient.

In many institutes, including Sigma, attendance and payment tracking are still done manually using paper records or spreadsheets. This often leads to common problems such as human errors, missing data, and wasted time. To solve these issues, this system introduces an automated solution using QR code technology.

In this system, each student registered at Sigma Institute is assigned a unique QR code. By simply scanning this code, attendance can be marked quickly and accurately for each class. Since students may attend multiple classes, the system keeps separate attendance records for every class and generates clear attendance reports for administrators to monitor student participation.

Along with attendance management, the system also includes a fee management module. Monthly fees can be calculated based on class settings or attendance rules. Administrators can easily generate payment reports in PDF format, and students receive receipts once payments are completed.

Overall, this system helps Sigma Institute reduce manual work, improve data accuracy, and manage students more effectively through a modern web-based platform using PHP and MySQL.

Background and Motivation

At Sigma Institute, managing student attendance and fee payments is an important daily task. However, most of these activities are still done manually using paper records or basic spreadsheets. This traditional method takes a lot of time and can easily lead to mistakes such as missing attendance, incorrect data entry, or lost records.

Because of these problems, staff often find it difficult to keep accurate and up-to-date information about students. It also becomes harder to track attendance properly, especially when students attend multiple classes. Fee collection and report generation can also be slow and confusing when done manually.

This situation creates a need for a better and more reliable system that can simplify these processes. That is the main motivation behind developing this QR code–based Student Management and Attendance System for Sigma Institute. By using modern web technology and QR code scanning, the system aims to make attendance marking faster, reduce human errors, and improve data accuracy.

In addition, automating fee management and report generation helps reduce the workload of administrative staff and improves overall efficiency. This system is designed to make daily operations smoother, more organized, and more reliable for both staff and students at Sigma Institute.

Problem in Brief

At Sigma Institute, student attendance and fee records are still managed using manual methods. This creates several practical problems in daily operations. Staff need to record attendance by hand, which takes time and can easily lead to mistakes or missing information. When students attend multiple classes, it becomes even more difficult to keep accurate records for each class.

Fee tracking is also done manually, so calculating payments and preparing reports can be slow and sometimes confusing. Because of this, there is a higher chance of errors, data loss, and delays in updating student information.

Overall, the current manual system is not efficient enough to handle the growing number of students and classes at Sigma Institute. This is why a more automated and reliable system is needed.

Aims and Objectives

Aim

The main aim of this project is to develop a QR code–based Student Management and Attendance System for Sigma Institute to improve the efficiency, accuracy, and reliability of managing student attendance and fee payments.

Objectives

- To develop a system to **manage student registration and store student details securely**.
- To **generate unique QR codes for each student** to simplify attendance marking.
- To **automate the attendance marking process** by using QR code scanning.
- To **record and manage attendance separately for each class**.
- To **calculate and manage student fee payments** based on class or attendance rules.
- To **generate attendance and payment reports in PDF format** for administrative use.
- To **provide payment receipts automatically** after successful transactions.
- To **reduce manual work and improve accuracy** in student and class management at Sigma Institute.

Proposed Solution

To solve the problems faced at Sigma Institute, a QR code–based Student Management and Attendance System is proposed. This system is designed as a web-based application that helps automate and simplify the way student records, attendance, and fee payments are managed.

In the proposed system, each student will be registered and assigned a unique QR code. Students only need to scan their QR code to mark attendance, which makes the process fast, simple, and accurate. The system will automatically record attendance for each class separately and store the data in a central database for easy access.

For fee management, the system will automatically calculate student payments based on class settings or defined rules. Administrators will be able to generate clear reports in PDF format, and students will receive receipts after completing payments.

Overall, this solution will reduce manual work, minimize errors, and improve the efficiency of daily operations at Sigma Institute by providing a modern and reliable digital system.

Resource Requirements

To develop and implement the QR code–based Student Management and Attendance System for Sigma Institute, several resources are required to ensure the system works efficiently and reliably.

First, a basic hardware setup is needed. This includes a computer or laptop for development and system administration, along with a smartphone or QR code scanner for marking attendance. A stable internet connection is also important since the system will be web-based.

Second, the system requires suitable software tools. The backend will be developed using **PHP** with a **MySQL** database to store and manage data. For the frontend, standard web technologies such as **HTML**, **CSS**, and **JavaScript** will be used to create a user-friendly interface. A local server environment like **XAMPP** can be used during development and testing.

Third, development tools such as a code editor (for example, VS Code) and a web browser for testing are necessary. In addition, PDF generation libraries and QR code generation libraries will be required to support key system functions.

Finally, human resources are also important. A developer is needed to design, build, and maintain the system, while administrative staff from Sigma Institute will help by providing requirements and testing feedback.

Overall, these resources will support the successful development and smooth operation of the system.

PROJECT SHEDULE

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